AIPE: context-aware replenishment policy selection

From heterogeneous demand to decisions for each store–item series

1 Characterize demand

Store A · item 1 — **Regular**

Store B · item 1 — **Intermittent**

Store C · item 2 — **Spikes**

Volume · variability
intermittency · trend

2 Select with AIPE

Demand features and observed profile

Portfolio of candidate policies
Fit on training · assess on validation

Two PSE approaches

Profile-based PSE

Given profile

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Profile-to-policy association

or

Predictive PSE

Series features
+ optional profile

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Model predicts the policy

Recommended policy for each store–item series

Selection before observing the test set

3 Recommend and test

Illustrative recommendations

- **Series A → policy A**
- **Series B → policy C**
- **Series C → policy B**

↓

When and how much to order

Held-out test evaluation
Cost · fill rate
selection stability

Compare against global selection

Adaptive Inventory Policy Engine

The same policy can perform differently across series.
AIPE enables evaluation of when contextual selection adds value.